

Dual-Stream Semantic Guidance with Prototype Anchor Calibration for Source-Fully-Free Adaptation of Vision-Language Models

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Abstract—Source-Fully-Free Domain Adaptation (SFF-DA) has emerged as a strategic paradigm to adapt Vision-Language Models (VLMs) without any access to source data or task-specific source models. However, we identify a critical *Dual Semantic Drift* that hinders this process: static drift arising from the rigidity of fixed class embeddings, and dynamic drift stemming from the divergence of generated captions, causing severe semantic misalignment that intensifies the stability-plasticity dilemma.

To address this, we propose DSSG (Dual-Stream Semantic Guidance), an end-to-end framework that reconciles fine-grained plasticity with global stability. Our core contribution is the Dual Semantic Guidance (DSG) module, which integrates a caption stream for domain-specific knowledge with a class-anchor stream to anchor global categorical consistency. Furthermore, a Dynamic Cross-Modal Knowledge Distillation (CMKD) module is introduced to leverage the evolving teacher distribution for calibrating teacher-student consistency. Building upon DSSG, we further introduce Prototype Anchor Calibration (PAC), yielding DSSG-PAC, which periodically calibrates prototype anchors and caches them until the next calibration. This design reduces redundant text-side computation while preserving the adaptability of class guidance to the evolving text space. We further establish SFF-DA risk bounds that relate student risk to semantic-teacher quality and teacher-student discrepancy. Extensive experiments demonstrate that DSSG consistently outperforms current state-of-the-art methods across multiple benchmarks, while DSSG-PAC largely preserves its adaptation performance with 18.9% lower total adaptation time. The code is available at <https://github.com/mrmenand/DSSG>.

Index Terms—Source-Fully-Free Domain Adaptation, Vision-Language Models, Cross-Modal Knowledge Distillation

I. INTRODUCTION

TRADITIONAL unimodal source-free domain adaptation (SFDA) follows a two-stage pipeline of source-domain pre-training and target-domain adaptation. While effective, this paradigm heavily relies on task-specific source priors and is restricted by practical deployment bottlenecks, such as data privacy concerns and model black-box constraints (e.g., API-only access). Recently, Vision-Language Models (VLMs) represented by CLIP, pre-trained on large-scale image-text pairs, exhibit strong zero-shot generalization, serving as generalized source models rich in transferable knowledge. To eliminate the dependency on task-specific source training, we focus on Source-Fully-Free Domain Adaptation (SFF-DA)—also known as Unsupervised Fine-Tuning (UFT) [1]. We directly leverage VLMs for adaptation on unlabeled target domains

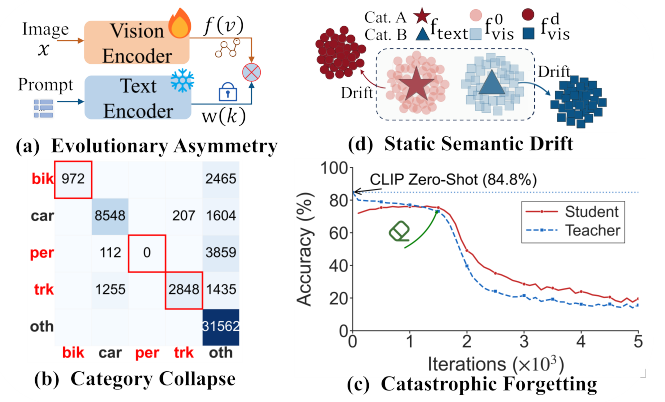


Fig. 1. Challenges of directly fine-tuning CLIP in SFF-DA. It can induce (b) Category Collapse and (c) Catastrophic Forgetting, while visual adaptation with fixed text embeddings causes (a) Evolutionary Asymmetry and (d) Static Semantic Drift. Here, f_0/d denote the initial and drifted visual features, respectively.

without accessing any source data or task-specific source models, aiming to transfer their inherent universal knowledge to domain-specific contexts under distribution shifts.

While parameter-efficient methods (e.g., Prompt Learning [2], Adapters [3]) preserve VLM priors by freezing backbones, their limited learnable parameters hinder the acquisition of intricate domain-specific knowledge. In contrast, directly fine-tuning foundation models in SFF-DA encounters a twofold challenge: (i) *error accumulation and catastrophic forgetting* (Fig. 1(b,c)), where the lack of source guidance accumulates pseudo-label errors, exacerbating confirmation bias and category collapse, thereby corrupting large-scale pre-trained knowledge; and (ii) *evolutionary asymmetry and static semantic drift* (Fig. 1(a, d)), where the vision encoder evolves toward a domain-specific manifold while template-derived embeddings w_k remain fixed in the frozen text space, inducing a spatiotemporal misalignment that undermines cross-modal consistency, termed static semantic drift.

Existing CLIP-based SFDA methods generally follow two paradigms. First, CLIP-assisted approaches (e.g., DIFO and ProDe [4], [5]) employ CLIP as an external teacher to calibrate task-specific source models. However, their reliance on pre-trained source weights limits their applicability to SFF-DA. Second, unsupervised fine-tuning approaches adapt VLMs by optimizing prompt variables [6], fine-tuning the vision encoder [7], or jointly tuning the vision and text encoders [8]. Although encoder updates improve target-domain adaptability, they may also compromise the generalizable knowledge acquired during pre-training. Together, these limitations expose

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a fundamental stability–plasticity dilemma: prompt-based tuning provides insufficient plasticity to capture target-specific distributions, whereas encoder-based tuning risks damaging pre-trained knowledge.

Motivation: Drawing inspiration from ImCapDA [9], we leverage VLM-generated captions as instance-level textual guidance and align them with target images through contrastive learning, enabling the vision and text encoders to jointly acquire target-domain semantics while facilitating cross-modal alignment. However, the absence of source supervision in the SFF-DA setting leaves the open-vocabulary semantics introduced by generated captions unconstrained, making caption-guided adaptation prone to dynamic semantic drift. Specifically, captions may emphasize non-discriminative instance details, background or out-of-distribution semantics, or cues from other task classes, reducing their reliability as category-discriminative guidance. Meanwhile, the fixed text class embeddings w_k remain anchored in the original text space and cannot track the evolving visual feature space, resulting in static semantic drift. Together, caption-induced dynamic semantic drift and static semantic drift from fixed class embeddings constitute the *Dual Semantic Drift* dilemma, making it difficult to acquire domain-specific knowledge while preserving global categorical consistency.

To address this, we present the Dual-Stream Semantic Guidance (DSSG) framework. First, the DSG module integrates a caption stream and a class-anchor stream to provide complementary semantic guidance: captions supply instance-specific context to mitigate forgetting, while class anchors preserve intrinsic category semantics. Second, the Dynamic CMKD module leverages the evolving teacher distribution through self-distillation to enforce teacher–student consistency. Together, these modules promote cross-modal alignment while preserving global categorical consistency. Building on DSSG, we introduce Prototype Anchor Calibration (PAC), yielding DSSG-PAC, which replaces iteration-wise class-prompt encoding with periodic prototype calibration and caching to reduce text-side training overhead while preserving adaptive category guidance.

Our main contributions are summarized as follows:

1. We identify Dual Semantic Drift in SFF-DA, arising from static class embeddings and unconstrained caption semantics, and establish risk bounds that relate student risk to semantic-teacher quality and teacher–student discrepancy.
2. We propose DSSG, which integrates dual-stream semantic guidance with Dynamic CMKD to enable caption-guided adaptation while preserving global categorical consistency.
3. We extend DSSG with Prototype Anchor Calibration, yielding DSSG-PAC, which replaces iteration-wise class-prompt encoding with periodic prototype calibration to reduce training overhead.
4. Extensive experiments demonstrate that DSSG achieves new state-of-the-art performance across multiple benchmarks, while DSSG-PAC largely preserves its accuracy with 18.9% lower total adaptation time.

II. RELATED WORK

Unimodal Source-Free Domain Adaptation. Unimodal SFDA methods typically rely solely on visual representations and leverage task-specific source models for self-training on unlabeled target data. SHOT performs information maximization and pseudo-label refinement [10], while AaD exploits neighborhood structure to improve target discriminability [11]. Subsequent methods improve adaptation robustness through domain-invariant parameter mining [12], target prediction distribution search [13], advanced data augmentation [14], and reliable pseudo-label selection [15]. More recently, RGV-SFDA analyzes SFDA from a target-risk perspective [16], whereas UCon-SFDA focuses on uncertainty control during adaptation [17]. However, unlike unimodal SFDA that adapts task-specific visual source models, SFF-DA directly adapts pre-trained VLMs without source data or task-specific source models.

Vision-Language Model Assisted SFDA. Recent SFDA methods introduce CLIP as an external multimodal teacher to guide source-pretrained visual models, mainly through two paradigms: (1) *Prediction-Level Calibration*: Co-learn++ [18] and DPC [19] ensemble zero-shot priors within dual-branch frameworks, while ProDe [5] utilizes CLIP as a noisy agent for probabilistic denoising. (2) *Structural-Level Alignment*: DIFO [4] maximizes mutual information for alignment, while MMGA [20] and DTKI [21] enforce topological consistency to anchor features to semantic prototypes. However, these methods still rely on task-specific source models and therefore do not satisfy the SFF-DA setting.

Multimodal Source-Fully-Free Domain Adaptation (SFF-DA). Existing works mainly follow three paradigms: (1) *Prompt Tuning*, which optimizes learnable prompts with frozen backbones (e.g., UPL [22], TFUP-T [6]); (2) *Visual Tuning*, which fine-tunes only the visual encoder to align with fixed text class embeddings (e.g., LaFTer [7], DPA [23]); and (3) *Joint Fine-tuning*, which optimizes both visual and text encoders for cross-modal alignment (e.g., ReCLIP [8], POUF [24], ImCapDA [9]). While prompt-based adaptation provides limited adaptability, visual tuning with fixed class embeddings can induce static semantic drift, whereas caption-guided joint fine-tuning may introduce dynamic semantic drift through unconstrained open-vocabulary semantics.

III. DSSG FRAMEWORK

We propose DSSG to address the *Dual Semantic Drift* dilemma in SFF-DA. It integrates VLM-generated captions, the DSG module, and Dynamic CMKD, while PAC extends the framework to DSSG-PAC by replacing iteration-wise class-prompt encoding with periodic prototype calibration.

A. Preliminaries and Problem Definition

SFF-DA Setting: Given an unlabeled target dataset $\mathcal{D}_t = \{x_i\}_{i=1}^{N_t}$, SFF-DA adapts a pre-trained vision-language model Θ_{clip} via unsupervised fine-tuning, without access to either source data $\mathcal{D}_s$ or source model Θ_s .

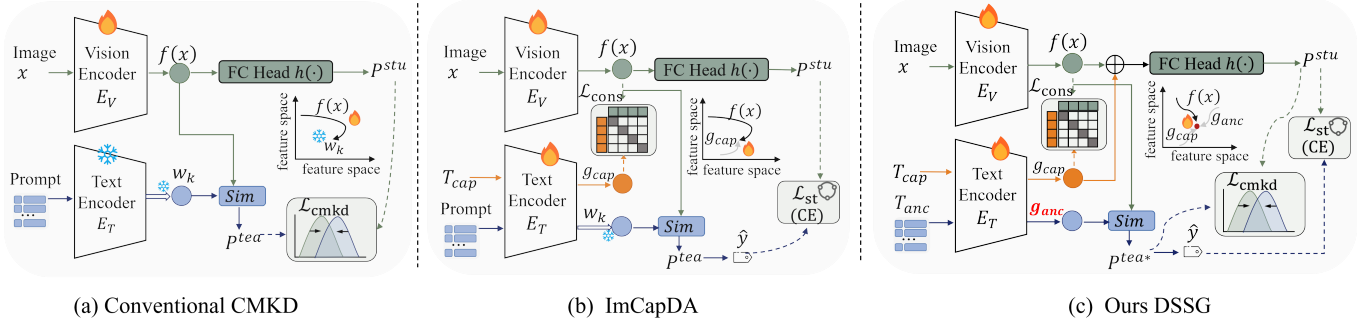


Fig. 2. Illustration of (a) Conventional CMKD, (b) ImCapDA, and (c) Our DSSG framework.

Revisiting CLIP: CLIP comprises a visual encoder $E_V(\cdot; \theta_V)$ and a text encoder $E_T(\cdot; \theta_T)$ that are jointly pre-trained via large-scale contrastive learning to align visual and textual representations in a shared feature space. During inference, given a category set $\mathcal{C} = \{c_k\}_{k=1}^K$, fixed class embeddings $w_k = E_T(t_k)$ are computed from prompt templates t_k (e.g., “a photo of a c_k ”). For an image x_i with feature $f_i = E_V(x_i)$, the prediction probability is defined as:

$$p(y_i = k | x_i) = \frac{\exp(\text{sim}(f_i, w_k) / \tau)}{\sum_{j=1}^K \exp(\text{sim}(f_i, w_j) / \tau)}, \quad (1)$$

where $\text{sim}(\cdot, \cdot)$ is the cosine similarity and τ is the temperature.

B. Data Priors: Generative Instance-level Captions

In SFF-DA, directly fine-tuning CLIP induces evolutionary asymmetry between visual features and fixed textual class embeddings w_k , resulting in static semantic drift. Although re-encoding class prompts at each iteration allows gradients to update the text encoder, such category-level guidance alone lacks instance-level target semantics and can suffer from catastrophic forgetting. Inspired by ImCapDA [9], we leverage VLM-generated captions to provide target-specific guidance for joint vision–text adaptation. We use the pre-generated captions released by ImCapDA, with each target image $x_i \in \mathcal{D}_t$ paired with an instance-level caption $T_{\text{cap},i}$. The caption generation process is formulated as:

$$T_{\text{cap},i} = \text{VLM}(x_i), \quad (2)$$

where the captions used in our experiments were generated by a frozen BLIP-3 model [25] with the prompt “Describe what this image is in one sentence.” Other VLMs can also be used as the caption generator. The generated captions $T_{\text{cap},i}$ provide instance-level target-domain semantics, including object attributes and contextual information. Through image–caption contrastive alignment, gradients are propagated to $E_T(\cdot; \theta_T)$, enabling the text encoder to acquire domain-specific knowledge alongside visual adaptation. This target-aware textual guidance helps alleviate catastrophic forgetting, a key challenge in SFF-DA without source guidance.

C. Dual-Stream Guidance

While T_{cap} introduces target-domain context, its unconstrained semantics can induce dynamic semantic drift, which,

together with the aforementioned static semantic drift, constitutes the *Dual Semantic Drift* dilemma. Unlike prior paradigms based on either fixed prompt templates [26] or caption-only guidance [9], DSG integrates a caption stream and a class-anchor stream in parallel. Specifically, for a target image x_i , its visual representation $f_i = E_V(x_i; \theta_V)$ is aligned with two complementary textual representations encoded by the shared text encoder $E_T(\cdot; \theta_T)$:

$$g_{\text{anc}} = E_T(T_{\text{anc}}; \theta_T), \quad g_{\text{cap},i} = E_T(T_{\text{cap},i}; \theta_T). \quad (3)$$

Here, $T_{\text{anc}} = \{t_k\}_{k=1}^K$ denotes the class prompt set, which is re-encoded at each training iteration by the current text encoder to obtain g_{anc} . The resulting g_{anc} preserves category semantics to mitigate dynamic semantic drift, whereas $g_{\text{cap},i}$ captures instance-specific target-domain semantics from VLM-generated descriptions to alleviate static semantic drift. Together, the two streams balance global semantic stability and local adaptation while maintaining cross-modal alignment.

To jointly fine-tune θ_V, θ_T , we adopt a self-distillation paradigm following prior works [9], [26], which employs a similarity-based teacher predictor and a linear student classifier $h(\cdot)$. Within this framework, we formulate objectives that exploit the adaptability of the caption stream while enforcing the semantic rigidity of the anchor stream:

Caption-guided Contrastive Alignment. To promote fine-grained cross-modal alignment, we employ a symmetric InfoNCE loss that maximizes the similarity of matched image–caption pairs $(f_i, g_{\text{cap},i})$ while minimizing similarities to mismatched pairs. The objective comprises an image-to-text loss $\mathcal{L}_{i2t}$ and a text-to-image loss $\mathcal{L}_{t2i}$:

$$\mathcal{L}_{i2t} = -\frac{1}{B} \sum_{i=1}^B \log \frac{\exp(\text{sim}(f_i, g_{\text{cap},i}) / \tau)}{\sum_{j=1}^B \exp(\text{sim}(f_i, g_{\text{cap},j}) / \tau)}, \quad (4)$$

$$\mathcal{L}_{t2i} = -\frac{1}{B} \sum_{i=1}^B \log \frac{\exp(\text{sim}(g_{\text{cap},i}, f_i) / \tau)}{\sum_{j=1}^B \exp(\text{sim}(g_{\text{cap},i}, f_j) / \tau)}. \quad (5)$$

where B denotes the mini-batch size. The resulting contrastive objective is $\mathcal{L}_{\text{con}} = 0.5(\mathcal{L}_{i2t} + \mathcal{L}_{t2i})$.

Dynamic Anchor-based Self-training. To address the *Dual Semantic Drift* described above, we update class anchors

$\{g_{\text{anc}}^k\}_{k=1}^K$ under the current θ_T to construct the dynamic teacher distribution:

$$\begin{aligned} P^{\text{tea}} &= \text{Softmax}(\text{sim}(f_i, w_k)/\tau) \quad (\text{Static}) \\ P^{\text{tea}*} &= \text{Softmax}(\text{sim}(f_i, g_{\text{anc}})/\tau) \quad (\text{Dynamic}), \end{aligned} \quad (6)$$

where P^{tea} provides dynamic guidance compared with the static baseline P^{tea} . Consequently, we derive pseudo-labels from the dynamic teacher: $\hat{y}_i = \arg \max_k P^{\text{tea}}(y = k|x_i)$. This dynamic anchor update is performed during training and introduces no additional cost at the inference stage.

Following ImCapDA [9], we obtain the student prediction P^{stu} by feeding the concatenated visual and caption features into a linear classifier $h(\cdot)$

$$P^{\text{stu}}(y|x_i) = \text{Softmax}(h([f_i; g_{\text{cap},i}])). \quad (7)$$

Based on the dynamic teacher prediction $P^{\text{tea}*}$, we formulate a confidence-thresholded self-training loss to transfer category-level anchor guidance to the student classifier. For each target sample, the teacher-derived pseudo-label $\hat{y}_i$ is used as supervision only when its prediction confidence exceeds γ :

$$\mathcal{L}_{\text{st}} = -\frac{1}{B} \sum_{i=1}^B \mathbb{I}(P^{\text{tea}*}(\hat{y}_i|x_i) \geq \gamma) \log P^{\text{stu}}(\hat{y}_i|x_i), \quad (8)$$

where $\mathbb{I}(\cdot)$ denotes the indicator function.

D. Dynamic Cross-Modal Knowledge Distillation

To harness the generalization capability of vision-language pre-training models, CMKD [26] leverages the static semantic guidance from frozen text encoders to transfer generic knowledge to the unlabeled target domain. However, its static teacher guidance cannot adapt to the evolving representation space during training, limiting teacher–student consistency. To bridge this gap, Dynamic CMKD leverages the dynamic teacher $P^{\text{tea}*}$ and student P^{stu} distributions obtained from the DSG module to enforce teacher–student consistency through a dynamic distillation objective.

Optimization Objective. To quantify the prediction agreement between the student and the dynamic teacher, we define a consistency coefficient $\text{coe} = \text{sg}(\exp(-\text{KL}(P^{\text{stu}} \| P^{\text{tea}})))$, where $\text{sg}(\cdot)$ denotes the stop-gradient operation. To mitigate the risk of overconfidence associated with standard entropy minimization, we formulate the Dynamic CMKD objective using Gini impurity:

$$\mathcal{L}_{\text{cmkd}}^* = \alpha \left[\underbrace{\text{coe} \cdot \mathcal{G}(P^{\text{stu}})}_{\mathcal{L}_{\text{task}}} + \underbrace{(1 - \text{coe}) \cdot \mathcal{G}(P^{\text{mix}})}_{\mathcal{L}_{\text{distill}}} \right] + \underbrace{\beta \cdot \mathcal{G}(P^{\text{tea}*})}_{\mathcal{L}_{\text{reg}}}, \quad (9)$$

where $\mathcal{G}(P) = 1 - |P|_2^2$ denotes the Gini impurity, and $P^{\text{mix}} = 0.5(P^{\text{stu}} + P^{\text{tea}})$ denotes the averaged student–teacher distribution. Specifically, when $\text{coe} \rightarrow 1$, $\mathcal{L}_{\text{task}}$ encourages confident student predictions. When $\text{coe} \rightarrow 0$, $\mathcal{L}_{\text{distill}}$ increasingly relies on P^{mix} , allowing the dynamic teacher prior to regularize the student and reduce prediction bias. Meanwhile,

$\mathcal{L}_{\text{reg}}$ promotes a sharper teacher distribution to prevent over-smoothing. We follow the original CMKD hyperparameter settings, with $\alpha = 0.25$ and $\beta = 0.025$.

E. Overall Optimization Objective

For end-to-end optimization, DSSG integrates the caption stream T_{cap} and class-anchor stream T_{anc} to jointly fine-tune the visual encoder θ_V and text encoder θ_T via the following total objective:

$$\mathcal{L}_{\text{total}} = \lambda_1 \mathcal{L}_{\text{con}} + \lambda_2 \mathcal{L}_{\text{st}} + \lambda_3 \mathcal{L}_{\text{cmkd}}^*, \quad (10)$$

where $\mathcal{L}_{\text{con}}$ promotes caption-guided cross-modal alignment, $\mathcal{L}_{\text{st}}$ performs confidence-thresholded self-training under dynamic teacher guidance, and $\mathcal{L}_{\text{cmkd}}^*$ encourages prediction consistency between the anchor-based dynamic teacher and the visual–caption student classifier. Together, these objectives mitigate the *Dual Semantic Drift*, while λ_1 , λ_2 , and λ_3 are trade-off hyperparameters.

Total Gradient. The visual and text encoders are jointly optimized through the aggregated gradients of the three objectives: $\nabla_{\theta_V, \theta_T} \mathcal{L}_{\text{total}} = \lambda_1 \nabla_{\theta_V, \theta_T} \mathcal{L}_{\text{con}} + \lambda_2 \nabla_{\theta_V, \theta_T} \mathcal{L}_{\text{st}} + \lambda_3 \nabla_{\theta_V, \theta_T} \mathcal{L}_{\text{cmkd}}^*$.

F. From DSSG to DSSG-PAC: Prototype Anchor Calibration

In DSSG, the class-anchor stream re-encodes all class prompts at each training iteration to obtain class text features from the current text encoder. Similarly, prompt-learning-based DA methods [2], [27] construct class prompts with learnable context token embeddings, which must still pass through the text encoder to produce the final class features. Their text-side encoding pipelines can be summarized as:

DSSG:

$$\begin{aligned} T_{\text{anc}} &\rightarrow \text{tokenized_prompts} \in \mathbb{N}^{K \times 77} \\ &\rightarrow \text{token_embedding} \in \mathbb{R}^{K \times 77 \times d_{\text{tok}}} \\ &\rightarrow E_T \rightarrow g_{\text{anc}} \in \mathbb{R}^{K \times d}, \end{aligned} \quad (11)$$

Prompt Learning:

$$\begin{aligned} \{[v_1, \dots, v_M, c_k]\}_{k=1}^K &\in \mathbb{R}^{K \times (M+1) \times d_{\text{tok}}} \\ \xrightarrow{\text{padding}_{77}} \text{token_embedding} &\in \mathbb{R}^{K \times 77 \times d_{\text{tok}}} \\ &\rightarrow E_T \rightarrow g_{\text{pl}} \in \mathbb{R}^{K \times d}, \end{aligned}$$

where $[v_1, \dots, v_M]$ denotes the shared learnable context embeddings, c_k denotes the fixed embedding corresponding to the k -th class name, and d_{tok} and d denote the token-embedding and text-feature dimensions, respectively. Although DSSG uses fixed class prompts whereas prompt-learning methods employ learnable contexts, both encode K class prompts through E_T at every iteration, incurring an iteration-level text-encoding cost that scales with the number of classes. We further instantiate prompt learning in DSSG as DSSG-PL, which replaces the fixed class prompts with learnable context embeddings.

Prototype Anchor Calibration. To avoid redundant encoding at every iteration, instead of optimizing class anchors

through prompt learning in the token-embedding space, we directly model the text-encoded class features as periodically calibrated prototype anchors in the text feature space, giving rise to DSSG with Prototype Anchor Calibration (DSSG-PAC). Let Δ_{cal} denote the calibration interval measured in epochs, and t_m the training iteration at which the m -th calibration is performed. At t_m , the text encoder generates the class prototypes using its current parameters as

$$g_{\text{pac}}^{(m)} = \text{sg} \left[E_T \left(T_{\text{anc}}; \theta_T^{(t_m)} \right) \right] \in \mathbb{R}^{K \times d}, \quad (12)$$

where $\theta_T^{(t_m)}$ denotes the current text-encoder parameters and $\text{sg}(\cdot)$ stops gradient propagation. These prototypes are fixed and reused as class anchors for $t_m \leq t < t_{m+1}$ without propagating gradients to E_T , whereas E_T remains optimized through the T_{cap} branch and its updated parameters are used to regenerate the prototypes at the next calibration. We set $\Delta_{\text{cal}} = 1$ by default, corresponding to one calibration per epoch, and evaluate other intervals in the ablation study.

Specifically, PAC replaces the g_{anc} generated at every iteration in the class-anchor stream with the periodically calibrated and reused $g_{\text{pac}}^{(m)}$, reducing text-encoding overhead while preserving the adaptability of the class prototypes to changes in the text feature space. Given the visual feature f_i of a target image x_i , the corresponding teacher distribution is

$$p_{\text{tea}}^*(y = k | x_i) = \frac{\exp \left(\cos \left(f_i, g_{\text{pac},k}^{(m)} \right) / \tau \right)}{\sum_{j=1}^K \exp \left(\cos \left(f_i, g_{\text{pac},j}^{(m)} \right) / \tau \right)}. \quad (13)$$

All other components and optimization objectives of DSSG remain unchanged.

For N_{ep} training epochs with I iterations per epoch, DSSG performs $N_{\text{ep}}I$ class-prompt encoding operations, each involving all K classes, resulting in a text-side encoding workload that scales with $N_{\text{ep}}IK$. Thus, the text-side overhead increases with both the target-domain size and the number of classes. In contrast, DSSG-PAC performs only approximately $\lceil N_{\text{ep}}/\Delta_{\text{cal}} \rceil$ such operations and maintains a $K \times d$ prototype matrix between calibrations.

Risk Analysis for Source-Fully-Free Adaptation

Unlike conventional UDA bounds that depend on source samples or a task-specific source hypothesis, neither is available in SFF-DA. We therefore characterize the target risk relative to the semantic teacher that is actually available during adaptation. Let $q(x) = P_{\text{tea}}^*(\cdot | x)$ and $p(x) = P_{\text{stu}}(\cdot | x)$, and let h_q and h_p be their induced top-1 classifiers. For the unknown target distribution Q , define the teacher risk $\eta_Q = \Pr_{(x,y) \sim Q}[h_q(x) \neq y]$ and the student risk $R_Q(h_p) = \Pr_{(x,y) \sim Q}[h_p(x) \neq y]$. Let $m_q(x) = q_{(1)}(x) - q_{(2)}(x)$ denote the teacher's top-1 probability margin and define

$$\Psi(p, q; x) = \begin{cases} \min \left\{ 1, \frac{2\|p(x) - q(x)\|_2^2}{m_q(x)^2} \right\}, & m_q(x) > 0, \\ 1, & m_q(x) = 0, \end{cases} \quad (14)$$

with $\mathcal{C}_Q(p, q) = \mathbb{E}_{x \sim Q_X} \Psi(p, q; x)$.

Theorem 1 (Certified SFF-DA risk bound). For any semantic-teacher and student distributions q and p ,

$$\Pr_{Q_X}[h_p(x) \neq h_q(x)] \leq \mathcal{C}_Q(p, q), \quad (15)$$

and

$$\max\{0, \eta_Q - \mathcal{C}_Q(p, q)\} \leq R_Q(h_p) \leq \min\{1, \eta_Q + \mathcal{C}_Q(p, q)\}. \quad (16)$$

The proof is provided in Appendix VI-E.

The bound separates the irreducible semantic quality of the dynamic teacher, η_Q , from a label-free, margin-normalized consistency certificate, $\mathcal{C}_Q$. In DSSG, the class-anchor stream is designed to reduce semantic drift in q , while Dynamic CMKD contains an explicitly disagreement-adaptive term proportional to $\|p - q\|_2^2$, which is aligned with the certificate in Eq. (14). We stress that a large teacher margin measures low predictive ambiguity rather than correctness; consequently, reduction of η_Q is an empirical property assessed through oracle pseudo-label analysis, not an assumption-free consequence of the bound.

IV. EXPERIMENTS

A. Experiments Setting

Datasets. Experiments are evaluated on four benchmarks: (i) *Office-31*, 4,110 images in 31 classes across three domains: Amazon (A), Webcam (W), and DSLR (D); (ii) *Office-Home*, 15,588 images in 65 classes across four domains: Art (A), Clipart (C), Product (P), and Real-World (R); (iii) *Mini-DomainNet*, a DomainNet subset with 140,006 images in 126 classes across four domains: Clipart (C), Painting (P), Real (R), and Sketch (S); (iv) *VisDA*, a synthetic-to-real benchmark with 55,388 target images across 12 classes.

Baselines. We evaluate our method against representative baselines from three categories: (i) *Unimodal Source-Free DA* (U), including SHOT [10], AaD [11], DIPE [12], DSiT [28], SF(DA)² [14], and TPDS [13]; (ii) *Multimodal Source-Free DA* (U+M), employing an external multimodal teacher to guide the source-supervised unimodal student, represented by DIFO-C [29], ProDe [5], MMGA [20], Co-Learn [18], DTKI [21], and DPC [19]; and (iii) *Multimodal Source-Fully-Free DA* (M), such as UPL [22], POUF [24], TFUP-T [6], ReCLIP [8], and ImCapSFDA [9].

Implementation Details. In the strict SFF-DA setting, target-domain labels are unavailable for model selection; therefore, we report Last Epoch accuracy as the primary evaluation metric. All experiments are conducted using three random seeds {2020, 2026, 2027}, and the averaged results are reported. Detailed experimental configurations are provided in Appendix VI-A.

B. Experimental Results

We compare DSSG and DSSG-PAC with representative baselines from unimodal (U), unimodal-student-with-multimodal-teacher (U+M), and directly fine-tuned multimodal (M) settings.

TABLE I
CLASSIFICATION ACCURACY (%) ON OFFICE-HOME. † DENOTES A MULTIMODAL TEACHER WITH ITS BACKBONE, “–” MEANS NOT REPORTED.

Methods	Venues	Prot.	Mark	C	P	R	A	P	R	A	C	R	A	C	P	Avg. Last	Avg. Best
				A			C			P			R				
<i>ResNet-50:</i>																	
SHOT [10]	ICML20	SF	U	68.0	67.4	73.3	57.1	54.9	58.8	78.1	78.2	84.3	81.5	78.1	82.2	71.8	–
SF(DA) ² [14]	ICLR24	SF	U	69.5	66.5	73.3	57.8	57.2	60.2	80.2	79.2	83.8	81.5	79.4	82.1	72.6	–
TIGM [15]	CVPR25	SF	U	69.3	68.1	74.6	61.2	58.8	62.4	80.9	81.2	85.7	82.7	81.4	83.4	74.1	–
DIFO-V [†] _{B/32} [29]	CVPR24	SF	U+M	82.5	80.9	83.4	70.6	70.1	70.5	90.6	90.6	91.2	88.8	88.8	88.9	83.1	–
ProDe-V [†] _{B/32} [5]	ICLR25	SF	U+M	82.5	82.5	83.0	72.7	72.5	72.6	92.3	91.5	92.2	90.5	90.7	90.8	84.5	–
MMGA [†] _{R/50} [20]	PR25	SF	U+M	77.2	75.9	79.4	65.8	65.3	66.1	84.5	86.9	88.7	85.9	86.1	86.1	79.0	–
Co-learn [†] _{L/14} [18]	IJCV25	SF	U+M	77.1	76.6	82.0	77.2	72.5	79.6	90.4	88.1	93.0	91.0	90.0	90.1	84.0	–
DTKI [†] _{B/32} [21]	IPM26	SF	U+M	82.3	82.8	82.7	72.0	71.1	71.7	91.7	91.9	91.6	90.0	90.0	90.3	84.0	–
CLIP-RN50 [30]	ICML21	SFF	M	71.5			54.5			80.0			82.3			72.1	–
ImCapSFDA [9]	ESWA26	SFF	M	81.6 ± 0.11			76.7 ± 0.08			88.6 ± 0.08			89.8 ± 0.14			84.2 ± 0.04	84.3 ± 0.05
DSSG	Ours	SFF	M	84.5 ± 0.20			83.7 ± 0.31			94.5 ± 0.20			92.4 ± 0.11			88.8 ± 0.06	89.2 ± 0.05
DSSG-PAC	Ours	SFF	M	84.5 ± 0.35			83.5 ± 0.27			94.2 ± 0.11			92.4 ± 0.15			88.7 ± 0.09	89.1 ± 0.11
<i>ViT-B/16:</i>																	
SHOT [10]	ICML20	SF	U	76.6	76.3	80.4	67.1	65.3	66.7	83.5	83.4	83.4	85.5	83.7	85.3	78.1	–
DIPE [12]	CVPR22	SF	U	77.1	75.3	81.6	66.0	63.3	67.7	80.8	83.5	89.6	85.6	83.4	85.1	78.2	–
DSiT [28]	ICCV23	SF	U	80.7	77.9	82.4	69.2	67.9	68.3	83.5	86.1	89.8	87.3	86.2	86.6	80.5	–
DPC [†] _{B/16} [19]	IJCAI24	SF	U+M	86.3	86.6	87.6	71.1	74.1	75.0	87.1	90.9	91.8	91.3	91.6	91.8	85.4	–
CLIP-ViT/B-16 [30]	ICML21	SFF	M	82.3			71.3			89.2			89.9			83.2	–
UPL [22]	ArXiv22	SFF	M	83.3			67.7			91.5			90.7			83.3	–
POUF [24]	ICML23	SFF	M	86.2			73.8			92.7			91.7			86.1	–
TFUP-T [6]	ArXiv24	SFF	M	86.0			74.2			93.1			91.7			86.3	–
ReCLIP [8]	WACV24	SFF	M	86.1			76.0			93.9			92.0			87.0	–
ImCapSFDA [9]	ESWA26	SFF	M	87.5 ± 0.05			87.0 ± 0.05			94.4 ± 0.01			93.9 ± 0.06			90.7 ± 0.03	90.8 ± 0.01
DSSG	Ours	SFF	M	90.0 ± 0.04			89.2 ± 0.15			97.3 ± 0.05			94.6 ± 0.09			92.8 ± 0.02	92.8 ± 0.01
DSSG-PAC	Ours	SFF	M	90.1 ± 0.08			89.2 ± 0.12			97.3 ± 0.10			94.6 ± 0.01			92.8 ± 0.02	92.8 ± 0.01

TABLE II
CLASSIFICATION ACCURACY (%) ON OFFICE-31. † DENOTES A MULTIMODAL TEACHER WITH ITS BACKBONE, “–” MEANS NOT REPORTED.

Methods	Prot.	Mark	D	W	A	W	A	D	Avg. Last	Avg. Best
			A		D		W			
<i>ResNet-50:</i>										
SHOT [10]	SF	U	74.7	77.8	94.0	99.8	90.1	98.4	88.6	–
SF(DA) ² [14]	SF	U	75.7	76.8	95.8	99.8	92.1	99.0	89.9	–
DIFO-V [†] _{B/32} [29]	SF	U+M	83.0	83.2	97.2	98.8	95.5	97.2	92.5	–
ProDe-V [†] _{B/32} [5]	SF	U+M	83.1	82.5	96.8	99.8	96.4	97.0	92.6	–
DTKI [†] _{B/32} [21]	SF	U+M	83.2	83.1	98.2	99.2	95.1	98.8	92.9	–
CLIP-RN50 [30]	SFF	M	73.2		72.5		70.1		71.9	–
ImCapSFDA [9]	SFF	M	77.8±0.34		88.0±0.40		86.2±0.08		84.0±0.04	84.2±0.09
DSSG	SFF	M	83.9±0.41		93.0±1.25		94.0±0.38		90.3±0.67	91.0±0.17
DSSG-PAC	SFF	M	83.5±0.32		92.0±1.23		93.5±0.82		89.7±0.49	90.4±0.16
<i>ViT-B/16:</i>										
SHOT [10]	SF	U	79.4	80.2	95.3	100.0	94.3	99.0	91.4	–
DIPE [12]	SF	U	77.5	77.1	94.8	100.0	95.5	98.5	90.5	–
DSiT [28]	SF	U	81.7	81.9	98.0	100.0	97.2	99.1	93.0	–
DPC [†] _{B/16} [19]	SF	U+M	83.9	83.7	96.9	100.0	97.0	98.2	93.3	–
CLIP-ViT-B/16 [30]	SFF	M	78.9		79.9		76.9		78.6	–
UPL [22]	SFF	M	81.4		82.6		83.6		82.5	–
POUF [24]	SFF	M	84.4		91.1		91.3		88.9	–
TFUP-T [6]	SFF	M	84.8		90.8		93.2		89.6	–
ImCapSFDA [9]	SFF	M	85.2±0.11		91.8±0.31		91.6±0.22		89.5±0.10	89.6±0.13
DSSG	SFF	M	87.6±0.13		98.4±0.00		97.2±0.00		94.4±0.04	94.8±0.06
DSSG-PAC	SFF	M	85.2±0.12		97.5±0.23		96.7±0.82		93.1±0.18	93.4±0.20

Results on Standard Benchmarks: Office-Home and Office-31. As shown in Table I, DSSG achieves **88.8%** and **92.8%** Avg. Last accuracy on Office-Home with ResNet-50 and ViT-B/16, outperforming ImCapSFDA by **4.6** and **2.1** percentage points, respectively. DSSG-PAC closely matches DSSG across both backbones. As shown in Table II, DSSG achieves **90.3%** and **94.4%** Avg. Last accuracy on Office-31 with ResNet-50 and ViT-B/16, respectively, consistently outperforming existing SFF baselines, while DSSG-PAC maintains competitive performance across both backbones.

Results on Large-Scale Benchmarks: Mini-DomainNet and VisDA. Table III and Table IV show that DSSG achieves **87.1%** and **89.67%** Avg. Last accuracy with ResNet, and

91.1% and **92.01%** with ViT-B/16 on Mini-DomainNet and VisDA, respectively. DSSG-PAC closely matches DSSG on both large-scale benchmarks.

Trade-offs and Fair Comparison: U+M vs. M Paradigms. Across the reported comparisons, DSSG consistently outperforms U+M baselines on both ResNet and ViT-B/16 backbones. These results indicate that U+M methods rely on source-trained initialization and external teacher guidance, whereas DSSG directly adapts a single VLM under the strict SFF setting without access to source models or external teachers. DSSG-PAC preserves this advantage while reducing the overhead of iteration-wise class-prompt encoding.

C. Ablation and Mechanism Analysis: DSSG and DSSG-PAC

Component Ablation of DSSG. To provide a stricter evaluation, we re-evaluate DSSG under the SFF-DA setting via component-wise removal with last-checkpoint accuracy, complementing the incremental best-checkpoint ablation in Appendix VI-B. Removing DSG consistently degrades performance, while removing Dynamic CMKD generally results in lower accuracy, validating the complementary roles of dual-stream guidance and dynamic distillation. Consistent with prior findings on catastrophic forgetting during direct CLIP fine-tuning [26], removing $\mathcal{L}_{con}$ causes substantial performance degradation, including a severe drop from **89.67%** to **10.35%** on VisDA with RN101. By enforcing instance-level, fine-grained image-caption alignment, $\mathcal{L}_{con}$ guides both encoders toward target-specific semantics while preserving the pretrained cross-modal representation space, thereby mitigating catastrophic forgetting. Removing $\mathcal{L}_{st}$ also causes sub-

TABLE III

CLASSIFICATION ACCURACY (%) ON MINI-DOMAINNET. † DENOTES A MULTIMODAL TEACHER WITH ITS BACKBONE, “–” MEANS NOT REPORTED.

Methods	Venues	Prot.	Mark	P	R	S	C	R	S	C	P	S	C	P	R	Avg. Last	Avg. Best
				C			P			R			S				
<i>ResNet-50:</i>																	
SHOT [10]	ICML20	SF	U	67.9	67.7	70.2	63.5	67.6	64.0	78.2	81.3	78.0	59.5	61.7	57.8	68.1	–
TPDS [13]	IJCV24	SF	U	65.6	66.4	68.6	62.9	67.0	64.3	77.1	79.0	75.3	59.8	61.5	68.2	67.1	–
DIFO-V [†] _{B/32} [29]	CVPR24	SF	U+M	80.0	80.8	80.5	76.6	77.3	76.7	87.2	87.4	87.3	74.9	75.6	75.5	80.0	–
ProDe-V [†] _{B/32} [5]	ICLR25	SF	U+M	85.0	85.5	85.5	83.2	83.1	83.4	92.4	92.3	92.4	79.0	79.3	79.1	85.0	–
MMGA [†] _{R/50} [20]	PR25	SF	U+M	74.6	75.9	–	–	76.1	75.6	–	88.3	–	69.9	–	70.9	75.9	–
Co-learn [†] _{L/14} [18]	IJCV25	SF	U+M	78.9	85.4	81.2	75.1	79.1	73.8	86.5	86.7	84.4	78.5	76.8	76.7	80.3	–
DTKI [†] _{B/32} [21]	IPM26	SF	U+M	82.4	80.7	82.3	77.5	78.7	78.9	88.9	88.0	88.3	76.1	76.9	74.6	81.1	–
CLIP-RN [30]	ICML21	SFF	M	68.4			69.3			86.2			65.9			72.5	–
ImCapSFDA [9]	ESWA26	SFF	M	84.2 ± 0.15			84.0 ± 0.03			91.9 ± 0.02			83.8 ± 0.06			86.0 ± 0.05	86.0 ± 0.06
DSSG	Ours	SFF	M	86.8 ± 0.00			84.5 ± 0.14			93.5 ± 0.06			84.2 ± 0.21			87.3 ± 0.08	87.3 ± 0.07
DSSG-PAC	Ours	SFF	M	86.5 ± 0.23			84.4 ± 0.12			93.4 ± 0.05			84.3 ± 0.17			87.2 ± 0.02	87.2 ± 0.01
<i>ViT-B/16:</i>																	
SHOT [10]	ICML20	SF	U	68.9	72.3	74.0	64.8	70.6	69.2	82.3	84.0	83.6	63.1	62.7	61.7	71.4	–
AaD [11]	NIPS22	SF	U	70.4	74.6	76.4	66.8	72.1	71.2	81.0	84.0	82.8	63.8	65.4	63.8	72.7	–
DPC [†] _{B/16} [19]	IJCAI24	SF	U+M	89.7	86.1	87.2	82.5	82.9	84.8	89.6	91.2	89.2	82.1	80.9	81.4	85.6	–
CLIP-ViT/B-16 [30]	ICML21	SFF	M	84.2			82.0			91.5			80.9			84.7	–
ImCapSFDA [9]	ESWA26	SFF	M	90.5 ± 0.03			89.4 ± 0.04			94.2 ± 0.02			89.1 ± 0.06			90.8 ± 0.03	90.8 ± 0.03
DSSG	Ours	SFF	M	90.8 ± 0.07			89.5 ± 0.07			95.1 ± 0.01			89.0 ± 0.04			91.1 ± 0.02	91.1 ± 0.02
DSSG-PAC	Ours	SFF	M	90.8 ± 0.10			89.4 ± 0.03			95.1 ± 0.01			89.0 ± 0.02			91.1 ± 0.03	91.1 ± 0.03

TABLE IV

PER-CATEGORY CLASSIFICATION ACCURACY (%) ON VISDA. † DENOTES A MULTIMODAL TEACHER WITH ITS BACKBONE, “–” MEANS NOT REPORTED.

Methods	Venues	Prot.	Mark	Pln	Bcy	Bus	Car	Hor	Kni	Mcy	Per	Plt	Skt	Tm	Trk	Avg. Last	Avg. Best
<i>ResNet-101:</i>																	
SHOT [10]	ICML20	SF	U	94.3	88.5	80.1	57.3	93.1	94.9	80.7	80.3	91.5	89.1	86.3	58.2	82.9	–
SF(DA) ² [14]	ICLR24	SF	U	96.8	89.3	82.9	81.4	96.8	95.7	90.4	81.3	95.5	93.7	88.5	64.7	88.1	–
DIFO-V [†] _{B/32} [29]	CVPR24	SF	U+M	97.5	89.0	90.8	83.5	97.8	97.3	93.2	83.5	95.2	96.8	93.7	65.9	90.3	–
ProDe-V [†] _{B/32} [5]	ICLR25	SF	U+M	98.3	92.4	86.6	80.5	98.1	98.0	92.3	84.3	94.7	97.0	94.1	75.6	91.0	–
MMGA [†] _{R/50} [20]	PR25	SF	U+M	97.2	85.0	90.2	82.9	96.8	97.1	93.4	84.6	90.7	94.6	90.4	60.7	88.6	–
Co-learn [†] _{L/14} [18]	IJCV25	SF	U+M	98.9	93.2	81.0	83.0	98.6	98.8	95.7	84.8	94.8	97.3	95.1	41.6	88.6	–
DTKI [†] _{B/32} [21]	IPM26	SF	U+M	97.7	87.7	87.5	82.7	97.3	98.3	93.3	85.1	95.3	96.3	94.0	73.9	90.8	–
CLIP-RN101 [30]	ICML21	SFF	M	98.2	83.9	90.5	73.5	97.2	84.0	95.3	65.7	79.4	89.9	91.8	63.3	84.4	–
ImCapSFDA [9]	ESWA26	SFF	M	99.4	91.0	93.6	79.0	99.3	94.3	96.2	54.4	94.5	99.5	94.7	68.5	88.7 ± 0.18	88.7 ± 0.18
DSSG	Ours	SFF	M	99.3	91.6	93.1	77.8	99.2	93.8	95.7	69.7	92.2	99.5	94.4	69.8	89.7 ± 0.10	89.8 ± 0.02
DSSG-PAC	Ours	SFF	M	99.4	91.6	93.0	77.9	99.2	94.1	95.7	69.2	92.5	99.5	94.4	69.9	89.7 ± 0.17	89.8 ± 0.10
<i>ViT-B/16:</i>																	
CLIP-ViT/B-16 [30]	ICML21	SFF	M	99.3	91.7	93.9	74.3	98.4	94.3	90.3	78.2	78.3	97.3	95.2	64.8	88.0	–
ImCapSFDA [9]	ESWA26	SFF	M	99.8	93.4	93.0	78.2	99.6	98.8	97.3	79.1	94.4	98.9	96.3	73.0	91.8 ± 0.07	91.8 ± 0.06
DSSG	Ours	SFF	M	99.7	93.2	92.4	81.3	99.5	98.7	97.0	82.5	93.0	98.7	96.7	71.5	92.0 ± 0.03	92.0 ± 0.03
DSSG-PAC	Ours	SFF	M	99.7	93.3	92.4	81.3	99.5	98.8	97.0	82.4	93.1	98.6	96.7	71.4	92.0 ± 0.04	92.0 ± 0.04

stantial degradation, highlighting its importance in converting dynamic teacher guidance into task-discriminative supervision.

Ablation Study on DSG Module. To examine the roles and complementarity of the two semantic streams in DSG, we remove $\mathcal{L}_{\text{cmkd}}^*$ to exclude the effect of Dynamic CMKD. Without semantic guidance, direct CLIP fine-tuning suffers severe degradation with ResNet, indicating catastrophic forgetting, while ViT-B/16 does not exhibit comparable collapse. The anchor stream shows similar degradation with ResNet, indicating that category-level semantics without instance-specific target information are insufficient for stable adaptation. Adding the class-anchor stream further enhances overall performance, demonstrating the complementary roles of the two streams: captions provide target-specific plasticity, whereas class anchors preserve task-relevant categorical stability.

Analysis of Prototype Anchor Construction and Calibration. DSSG-PAC introduces two changes to the class-anchor stream: gradient detachment and periodic prototype

calibration. We introduce DSSG-StopGrad to disentangle these two factors by detaching class-anchor gradients while retaining iteration-wise recomputation with the current E_T . DSSG-StopGrad exhibits noticeable degradation relative to DSSG mainly on Office-31, while remaining comparable on Office-Home and VisDA, suggesting that class-anchor gradients have limited overall impact. This degradation is primarily observed in target domains with fewer samples, possibly due to their greater sensitivity to gradient variations. In contrast, DSSG-PAC performs comparably to DSSG-StopGrad across all settings, indicating that periodically calibrated prototype anchors provide sufficient class guidance in most settings without requiring iteration-wise recomputation or gradient propagation through the anchor branch.

To further examine class-anchor construction, DSSG-PL adopts learnable context embeddings while retaining iteration-wise text encoding, whereas DSSG-LearnProto directly learns class prototypes without E_T . DSSG-PL provides no consistent gains across datasets or backbones, indicating limited

TABLE V

COMPONENT ABLATION OF DSSG UNDER STRICT SFF-DA. DOMAIN VISDA RESULTS ARE REPORTED AS MEAN LAST-CHECKPOINT ACCURACY (%) $\pm$ STANDARD DEVIATION. PARENTHESES IN AVG. AND VISDA DENOTE CHANGES FROM DSSG IN PERCENTAGE POINTS.

Variant	CLIP-RN50										CLIP-RN101
	Office-31 (31 classes)					Office-Home (65 classes)					VisDA (12 classes)
	A	D	W	Avg.		A	C	P	R	Avg.	R
DSSG	83.93 $\pm$ 0.41	92.97 $\pm$ 1.25	93.96 $\pm$ 0.38	90.29		84.52 $\pm$ 0.20	83.67 $\pm$ 0.31	94.50 $\pm$ 0.20	92.44 $\pm$ 0.11	88.78	89.67 $\pm$ 0.10
w/o DSG	80.84 $\pm$ 0.24	86.88 $\pm$ 0.23	91.53 $\pm$ 1.72	86.42 (-3.87)		83.00 $\pm$ 0.46	80.61 $\pm$ 0.17	92.00 $\pm$ 0.08	91.74 $\pm$ 0.00	86.84 (-1.94)	88.79 $\pm$ 0.08 (-0.88)
w/o $\mathcal{L}_{cmkd}^*$	81.35 $\pm$ 0.16	89.49 $\pm$ 0.12	91.74 $\pm$ 0.07	87.53 (-2.76)		82.45 $\pm$ 0.37	82.23 $\pm$ 0.23	92.02 $\pm$ 0.25	91.40 $\pm$ 0.13	87.02 (-1.76)	89.68 $\pm$ 0.18 (+0.01)
w/o $\mathcal{L}_{con}$	75.97 $\pm$ 3.59	76.31 $\pm$ 1.57	74.72 $\pm$ 1.53	75.66 (-14.63)		69.34 $\pm$ 1.73	41.64 $\pm$ 4.63	89.75 $\pm$ 0.30	88.51 $\pm$ 0.30	72.31 (-16.47)	10.35 $\pm$ 0.23 (-79.32)
w/o $\mathcal{L}_{st}$	78.63 $\pm$ 0.22	89.69 $\pm$ 0.50	89.01 $\pm$ 1.74	85.78 (-4.51)		60.54 $\pm$ 1.34	55.24 $\pm$ 1.69	71.11 $\pm$ 1.21	70.48 $\pm$ 4.98	64.34 (-24.44)	70.34 $\pm$ 3.41 (-19.33)

Variant	CLIP-ViT-B/16										CLIP-ViT-B/16
	Office-31 (31 classes)					Office-Home (65 classes)					VisDA (12 classes)
	A	D	W	Avg.		A	C	P	R	Avg.	R
DSSG	87.55 $\pm$ 0.13	98.39 $\pm$ 0.00	97.23 $\pm$ 0.00	94.39		90.03 $\pm$ 0.04	89.20 $\pm$ 0.15	97.31 $\pm$ 0.05	94.64 $\pm$ 0.09	92.79	92.01 $\pm$ 0.03
w/o DSG	85.41 $\pm$ 0.07	91.77 $\pm$ 0.69	92.58 $\pm$ 0.21	89.92 (-4.47)		88.37 $\pm$ 0.06	88.08 $\pm$ 0.08	94.46 $\pm$ 0.15	94.24 $\pm$ 0.02	91.29 (-1.50)	91.79 $\pm$ 0.08 (-0.22)
w/o $\mathcal{L}_{cmkd}^*$	85.00 $\pm$ 0.13	91.77 $\pm$ 0.20	95.22 $\pm$ 0.00	90.66 (-3.73)		89.36 $\pm$ 0.09	88.05 $\pm$ 0.04	96.86 $\pm$ 0.01	94.38 $\pm$ 0.07	92.16 (-0.63)	92.00 $\pm$ 0.02 (-0.01)
w/o $\mathcal{L}_{con}$	85.50 $\pm$ 0.29	87.48 $\pm$ 0.42	89.77 $\pm$ 0.07	87.59 (-6.80)		86.72 $\pm$ 0.28	82.01 $\pm$ 0.85	94.17 $\pm$ 0.05	94.15 $\pm$ 0.09	89.26 (-3.53)	90.74 $\pm$ 0.06 (-1.27)
w/o $\mathcal{L}_{st}$	84.88 $\pm$ 0.07	92.30 $\pm$ 0.50	92.32 $\pm$ 0.65	89.84 (-4.55)		62.01 $\pm$ 0.39	63.96 $\pm$ 0.92	71.99 $\pm$ 0.57	69.79 $\pm$ 0.92	66.93 (-25.86)	86.84 $\pm$ 0.29 (-5.17)

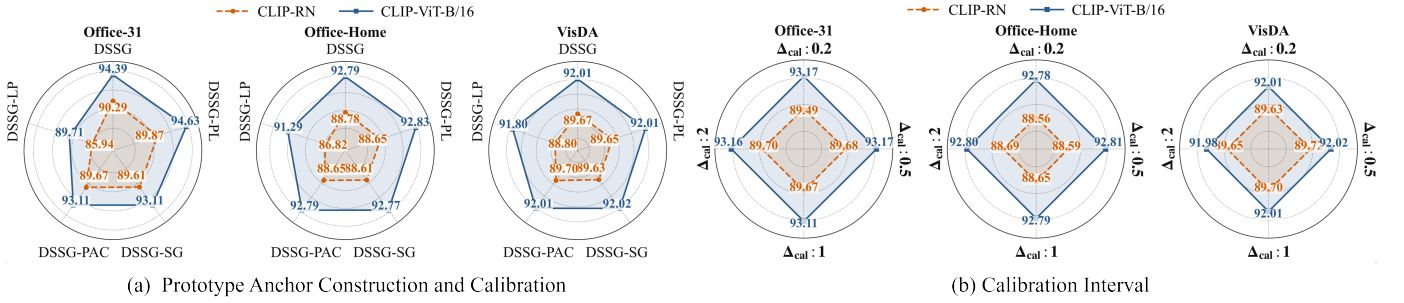


Fig. 3. Prototype anchor analysis of DSSG-PAC: (a) construction and calibration; (b) sensitivity to calibration interval Δ_{cal} .

TABLE VI

ABLATION STUDY OF THE DSG MODULE. $\ddagger$ INDICATES CATASTROPHIC FORGETTING.

Method	O-31	O-Home	MiniDN	VisDA	Avg.
CLIP-RN	71.90	72.10	72.50	84.40	75.23
FT w/o Stream $\ddagger$	35.74	8.00	6.14	10.87	15.19
Anchor Stream $\ddagger$	27.92	10.70	2.98	10.62	13.06
Caption Stream	83.99	84.18	85.97	88.71	85.71
Dual Stream (DSG)	87.53	87.02	85.91	89.68	87.54

CLIP-ViT-B/16	78.60	83.40	84.70	88.00	83.68
FT w/o Stream	84.84	85.68	88.20	90.58	87.33
Anchor Stream	85.09	86.25	88.46	90.62	87.61
Caption Stream	89.53	90.69	90.78	91.80	90.70
Dual Stream (DSG)	90.66	92.16	90.86	92.00	91.42

benefit from prompt learning once the text encoder has been adapted through caption guidance, while retaining the same encoding overhead. In contrast, DSSG-LearnProto consistently degrades performance, showing that unconstrained prototypes cannot replace text-encoder-conditioned category semantics. Additionally, other PAC variants based on momentum and residual correction are evaluated in Appendix VI-C.

Mechanism Analysis: Prototype Staleness and Teacher Prediction Consistency. We investigate whether the periodically reused prototype anchors $g_{pac}^{(m)}$ deviate from the class anchors $g_{anc}^{(t)}$ generated by the evolving text encoder, and whether the resulting anchor discrepancy affects teacher predictions.

For $t \in [t_m, t_{m+1})$, we quantify this anchor discrepancy as prototype staleness using their mean cosine distance, and record its maximum within each calibration interval:

$$S_t = 1 - \frac{1}{K} \sum_{k=1}^K \langle g_{anc,k}^{(t)}, g_{pac,k}^{(m)} \rangle, \quad (17)$$

$$S_m^{\max} = \max_{t_m \leq t < t_{m+1}} S_t.$$

To assess teacher prediction consistency, we compute the batch-averaged KL divergence between the teacher predictions induced by $g_{anc}^{(t)}$ and $g_{pac}^{(m)}$, together with its mean over each calibration interval:

$$D_t^{\text{KL}} = \frac{1}{B_t} \sum_{i=1}^{B_t} D_{\text{KL}}(P_{i,t}^{\text{tea,anc}} \| P_{i,t}^{\text{tea,pac}}), \quad (18)$$

$$\bar{D}_m^{\text{KL}} = \frac{1}{t_{m+1} - t_m} \sum_{t=t_m}^{t_{m+1}-1} D_t^{\text{KL}}.$$

Within each calibration interval, prototype staleness starts from zero and increases approximately monotonically. As shown in Fig. 4, maximum staleness and mean teacher KL rapidly decrease to below 10^{-4} and approximately 10^{-5} , respectively, demonstrating the rapid stabilization of class prototypes after early adaptation. Thus, periodically calibrated prototype anchors maintain teacher prediction consistency without requiring iteration-wise class-prompt encoding.

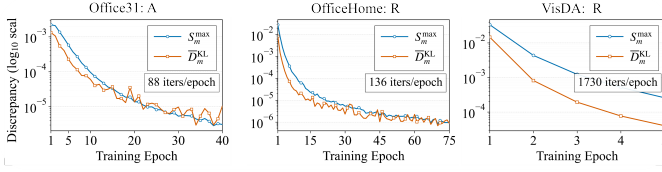


Fig. 4. Prototype staleness and teacher prediction discrepancy.

Effect of the Calibration Interval. As shown in Fig. 3, $\Delta_{\text{cal}} = 0.2, 0.5, 1$, and 2 yield comparable results, indicating that PAC is robust to variations in the calibration interval. This finding is consistent with the observed stabilization of prototype anchors after early adaptation. Considering the calibration overhead and VisDA’s five-epoch training schedule, we adopt $\Delta_{\text{cal}} = 1$ as the default.

D. Computational Efficiency Analysis

Training and Inference Efficiency Analysis. Table VII shows that DSSG adds no trainable parameters over ImCapSFDA but incurs adaptation overhead from iteration-wise class-prompt encoding. Periodic prototype calibration substantially reduces this cost in DSSG-PAC. At inference, DSSG computes class anchors only once, while DSSG-PAC directly uses the calibrated prototype anchors, leaving no additional downstream inference overhead. The cost of local caption generation is also reported, with batching substantially reducing the per-image latency.

TABLE VII
COMPUTATIONAL EFFICIENCY COMPARISON OF DSSG AND DSSG-PAC.

Model	Training			Inference		
	Trainable Params (M)	Iter Speed (it/s)	GPU Mem (MB)	Latency (ms)	GFLOPs	GPU Mem (MB)
CLIP-ViT-B/16	0.00	-	-	6.59	33.71	1228
ImCapSFDA	124.09	12.93	5356	6.63	39.53	1228
DSSG	124.09	12.09	5838	6.64	39.53	1228
DSSG-PAC	124.09	12.91	5570	6.63	39.53	1228
Local Captioner (e.g., BLIP-3)	Caption Generation 848 ($B=1$) / 203 ($B=12$) ms/img			GPU Memory 9842 MB		

Adaptation-Time Efficiency Analysis. As shown in Table VIII, DSSG-PL moderately reduces adaptation time by avoiding repeated prompt-to-token-embedding conversion, whereas DSSG-PAC achieves an average 18.9% reduction over DSSG and a runtime comparable to ImCapSFDA, further validating the efficiency of periodic prototype calibration.

E. Quantitative and Qualitative Analysis of Caption Semantics

Task-Relevant Proxies for the Semantic Quality of Captions. Without target labels, the quality of generated captions and their impact on classification cannot be directly assessed, motivating the use of task-relevant proxy metrics. Explicit and implicit class matching [9] quantify ground-truth class-name and synonym occurrences, respectively, while class ambiguity measures captions matching multiple task classes. We further match each caption to class texts by cosine similarity and

TABLE VIII
TOTAL ADAPTATION TIME (MIN) USING CLIP-ViT-B/16, INCLUDING BOTH TRAINING AND EVALUATION.

Method	CLIP-ViT-B/16								
	Office-31			Office-Home			VisDA	Overall	
	A	D	W	A	C	P	R	Val	Reduction (%)
ImCapSFDA	5.5	3.8	4.7	8.9	15.5	15.7	16.4	18.9	19.1
DSSG	6.5	4.2	5.8	11.8	20.5	20.8	21.4	21.4	-
DSSG-PL	6.0	4.1	5.5	11.0	19.2	19.5	20.2	20.3	5.7
DSSG-PAC	5.4	3.8	5.0	8.9	15.3	15.5	16.3	18.9	18.9

use the most similar class as the prediction to assess class discriminability.

Table IX shows that the ground-truth class name is explicitly matched in only a subset of captions, while implicit matching captures more synonymous class expressions, yet 16%–39% of captions still match multiple task classes, revealing intra-task class ambiguity and making direct class-term matching unreliable for standalone classification. Caption-to-class similarity prediction outperforms CLIP zero-shot on several domains but remains below caption-driven adaptation, suggesting that captions are more effective as complementary visual-semantic guidance than for classification alone.

TABLE IX
CAPTION QUALITY PROXIES AND CLASSIFICATION PERFORMANCE (%).

Metric / Method	Office-31			Office-Home				VisDA
	A	D	W	A	C	P	R	Val
Explicit matching rate	51.76	58.63	56.86	59.46	63.41	78.37	72.64	57.38
Implicit matching rate	58.47	71.08	69.43	70.37	75.58	86.96	81.09	62.86
Implicit ambiguity rate	16.05	20.48	20.25	22.58	33.86	38.66	32.64	19.78
Class-Name Similarity	77.32	79.72	82.01	77.75	78.97	91.71	84.71	77.47
Prompted-Class Similarity	77.32	83.53	83.02	78.66	79.93	93.20	87.26	79.74
CLIP-ViT-B/16	78.90	79.90	76.90	82.30	71.30	89.20	89.90	88.00
ImCapSFDA	85.19	91.84	91.57	87.52	86.99	94.36	93.89	91.75
DSSG (Ours)	87.55	98.39	97.23	90.03	89.20	97.31	94.64	92.03

Qualitative Analysis of Caption-Induced Semantic Drift.

The quantitative results reveal class matching and ambiguity but not the specific distracting semantics in captions. As illustrated in Fig. 5, captions may mix class-discriminative cues with background/OOD semantics or cues from other task classes, shifting their semantic focus away from the target category. Thus, caption quality depends not only on descriptive richness but also on whether the semantic focus remains task-relevant. DSSG therefore complements open-vocabulary caption guidance with class-anchor guidance to preserve category-level focus during adaptation.

Visualization Analysis. t-SNE visualizations in Fig. 6 show that DSSG yields more compact and better-separated target clusters than CLIP zero-shot across all four benchmarks, indicating improved target-domain discriminability.

F. Discussion

From Dual-Stream Semantic Guidance to Efficient Prototype Calibration. Our analyses reveal that captions provide instance-specific semantics but may also introduce intra-task ambiguity and semantic drift. DSSG balances semantic

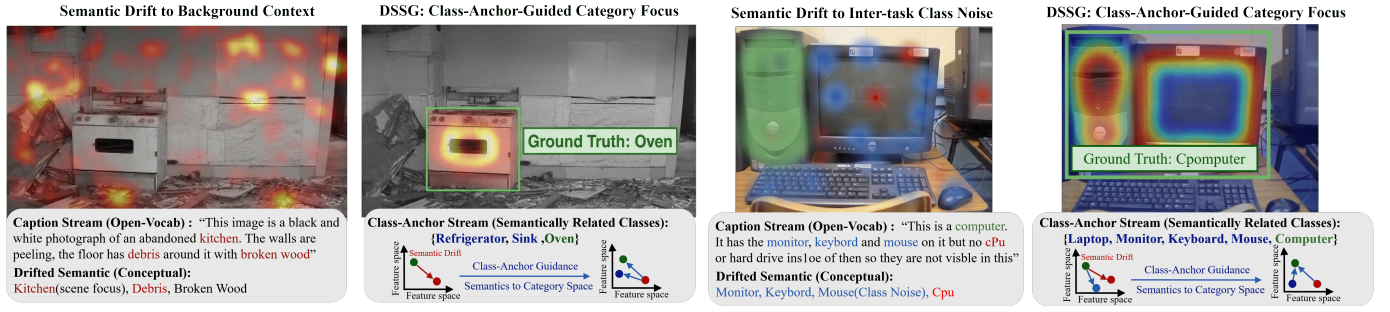


Fig. 5. Caption-induced semantic drift and class-anchor guidance. Green, blue, and red mark ground truth, intra-task class noise, and background/OOD noise.

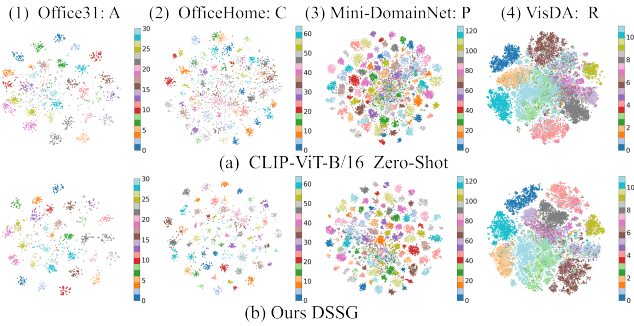


Fig. 6. t-SNE visualizations across four benchmarks.

plasticity and stability by coupling caption guidance with task-relevant class anchors, while Dynamic CMKD promotes teacher–student consistency. As the anchor space stabilizes during adaptation, DSSG-PAC periodically calibrates and reuses prototype anchors, reducing redundant text encoding while largely retaining the adaptive category guidance of DSSG.

Strict Source-Fully-Free Adaptation. DSSG operates under the strict SFF-DA protocol, requiring neither source data nor a task-specific source model. Unlike U methods that inherit source-supervised models and U+M methods that additionally employ external multimodal teachers, DSSG adapts a single pretrained VLM using only unlabeled target data and generated captions. Its competitive or superior results across these protocols demonstrate that generic vision–language knowledge can effectively compensate for unavailable source supervision.

Comparison with Zero-Shot MLLM Inference. Table X compares DSSG with the zero-shot MLLM results reported in [31], including LLaVA-v1.6 [32], InstructBLIP [33], and ShareGPT4V [34]. These methods perform VQA-based classification using the target image and all class names, followed by STS matching with *all-MiniLM-L6-v2* ($\sim 0.02B$). DSSG outperforms these MLLMs on average on Office-Home and MiniDomainNet and matches LLaVA-v1.6-34B on VisDA. Instead of repeatedly invoking a large MLLM for prediction, DSSG adapts a lightweight CLIP backbone (0.12B), while captions may be generated locally, obtained via APIs, or pre-computed offline.

Limitations and Future Work. DSSG remains dependent

TABLE X
COMPARISON WITH ZERO-SHOT MLLM INFERENCE

Method	Office-Home					MiniDomainNet					VisDA	
	A	C	P	R	Avg.	C	P	R	S	Avg.	Val	
LLaVA-v1.6-34B	87.0	78.3	93.7	89.5	87.1	85.5	84.4	91.0	83.7	86.2	92.1	
InstructBLIP-XXL-12B	82.2	82.0	91.6	88.8	86.2	86.7	82.5	89.0	83.0	85.3	86.7	
ShareGPT4V-13B	83.2	66.7	85.8	84.8	80.1	79.9	79.7	87.9	79.2	81.7	90.4	
DSSG-0.12B (BLIP-3, 4.1B)	90.1	89.4	97.4	94.6	92.9	91.1	83.7	94.9	87.6	89.3	92.1	

on caption quality and acquisition. Locally deployed captioners introduce computation, offline captions require storage, and online APIs incur latency and potential privacy concerns. Moreover, task-irrelevant or biased descriptions may still affect adaptation despite class-anchor regularization. Future work will explore uncertainty-aware caption refinement, lightweight captioning, and adaptive prototype calibration for more reliable and efficient semantic guidance.

V. CONCLUSION

We investigate strict Source-Fully-Free Domain Adaptation and identify Dual Semantic Drift arising from fixed class embeddings and unconstrained caption semantics. DSSG addresses these complementary failure modes by coupling instance-specific captions with dynamically updated class anchors, while Dynamic CMKD enforces consensus-aware teacher–student consistency. DSSG-PAC further replaces iteration-wise class-prompt encoding with periodic prototype calibration. Our theoretical analysis relates the student risk to semantic-teacher quality and margin-normalized teacher–student discrepancy, clarifying the conditions for reliable adaptation. Experiments across multiple benchmarks demonstrate strong performance under the strict SFF-DA protocol, while DSSG-PAC largely preserves accuracy with 20% lower training time. These results establish an effective accuracy–efficiency trade-off for source-independent adaptation of vision–language models.

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VI. APPENDIX MATERIAL

A. Experimental Configurations

Table XI summarizes the experimental configurations and hyperparameter settings for all benchmarks and backbones to facilitate reproducibility.

TABLE XI
TRAINING CONFIGURATIONS UNDER THE STRICT SFF-DA SETTING.

Parameter	Office-31	Office-Home	Mini-DomainNet	VisDA-2017
Common Settings	CLIP-RN50 (RN101 on VisDA) / ViT-B/16, WIT-400M SGD ($m = 0.9$, weight decay $5e-4$), Batch size 32, $\gamma = 0.9$ FC-Head LR: $1000 \times$ encoder LR; Seeds: {2020, 2026, 2027}; PyTorch, RTX 4090D			
Target Samples	A: 2,817 D: 498 W: 795	A: 2,427 C: 4,365 P: 4,439 R: 4,357	C: 18,523 P: 30,042 R: 69,622 S: 24,147	R: 55,388
Epochs	A: 40 D/W: 120	75	10	5
LR (RN)	$3e-6$	$3e-6$	$6e-7$	$5e-8$
LR (ViT)	$3e-6$	$3e-6$	$1.5e-6$	$3e-7$
$(\lambda_1, \lambda_2, \lambda_3)$	(0.5, 0.1, 0.5)	(1.0, 0.05, 0.2)	(0.3, 0.2, 0.4)	(1.0, 1.0, 0.1)

B. Incremental Ablation of DSSG under Best-Checkpoint Evaluation

Table XII complements the main-text component-removal ablation with an incremental best-checkpoint analysis. DSG (②) and static CMKD (③) independently improve the baseline and provide further gains when combined (④). Dynamic CMKD (⑥) consistently outperforms its static counterpart across datasets and backbones, while the degradation of ⑤ confirms the importance of caption-based contrastive alignment.

TABLE XII
ABLATION STUDY OF DSSG. ID ① DENOTES THE ImCapSFDA BASELINE. DSG DENOTES THE DUAL-STREAM GUIDING STRUCTURE.

CLIP-RN50															RN101		
ID	DSG	Loss	Office31(31 classes)				Office-Home(65 classes)					MiniDomainNet(126 classes)					VisDA(12 classes)
			A	D	W	Avg.	A	C	P	R	Avg.	C	P	R	S	Avg.	Real
①	-	$\mathcal{L}_{con} + \mathcal{L}_{st}$	78.13	88.55	86.04	84.24	81.87	77.25	88.92	89.95	84.50	83.17	77.62	90.95	78.89	82.66	88.95
②	✓	$\mathcal{L}_{con} + \mathcal{L}_{st}$	81.54	90.16	92.33	88.01	82.94	82.57	92.25	91.74	87.38	83.49	76.35	92.06	78.89	82.70	89.72
③	-	$\mathcal{L}_{con} + \mathcal{L}_{st} + \mathcal{L}_{cmkd}$	82.00	90.16	92.70	88.29	83.48	80.38	92.43	92.06	87.09	85.87	77.62	92.22	80.32	84.01	89.12
④	✓	$\mathcal{L}_{con} + \mathcal{L}_{st} + \mathcal{L}_{cmkd}$	83.95	91.77	93.58	89.77	85.25	83.96	94.30	92.66	89.04	84.92	76.35	92.70	80.48	83.61	89.60
⑤	✓	$\mathcal{L}_{st} + \mathcal{L}_{cmkd}^*$	84.06	81.12	82.64	83.35	83.07	69.19	91.24	87.80	82.83	84.76	76.67	91.75	76.51	82.42	84.07
⑥	✓	$\mathcal{L}_{con} + \mathcal{L}_{st} + \mathcal{L}_{cmkd}^*$	85.27	92.57	94.09	90.64	85.41	84.08	94.80	92.82	89.28	85.56	77.94	92.86	80.00	84.09	89.98
CLIP-ViT-B/16																	
ID	DSG	Loss	Office31(31 classes)				Office-Home(65 classes)					MiniDomainNet(126 classes)					VisDA(12 classes)
			A	D	W	Avg.	A	C	P	R	Avg.	C	P	R	S	Avg.	Real
①	-	$\mathcal{L}_{con} + \mathcal{L}_{st}$	85.20	92.17	93.08	90.15	87.76	87.06	94.50	94.29	90.90	90.16	83.33	94.29	86.83	88.65	91.75
②	✓	$\mathcal{L}_{con} + \mathcal{L}_{st}$	85.55	95.38	95.35	92.09	89.62	88.06	96.91	94.38	92.24	89.68	82.86	94.92	87.30	88.69	92.08
③	-	$\mathcal{L}_{con} + \mathcal{L}_{st} + \mathcal{L}_{cmkd}$	86.16	92.97	92.20	90.44	88.75	88.04	94.66	94.22	91.42	90.32	83.33	94.13	87.14	88.73	91.78
④	✓	$\mathcal{L}_{con} + \mathcal{L}_{st} + \mathcal{L}_{cmkd}$	86.01	96.18	94.97	92.39	90.07	89.21	97.25	94.58	92.78	91.11	83.17	94.92	87.62	89.21	92.07
⑤	✓	$\mathcal{L}_{st} + \mathcal{L}_{cmkd}^*$	87.50	87.95	91.19	88.88	87.80	82.96	94.32	94.15	89.81	90.16	81.75	94.92	86.19	88.26	90.68
⑥	✓	$\mathcal{L}_{con} + \mathcal{L}_{st} + \mathcal{L}_{cmkd}^*$	87.79	98.59	98.36	94.91	90.11	89.39	97.36	94.61	92.87	91.11	83.65	94.92	87.62	89.33	92.13

TABLE XIII
ABLATION STUDY ON PAC VARIANTS. RESULTS ARE REPORTED AS MEAN ACCURACY (%) $\pm$ STANDARD DEVIATION.

Variant	Office-31				Office-Home					VisDA-2017	
	A	D	W	Avg.	A	C	P	R	Avg.	R	
DSSG-PAC	85.16 $\pm$ 0.12	97.46 $\pm$ 0.23	96.73 $\pm$ 0.82	93.11 $\pm$ 0.18	90.06 $\pm$ 0.08	89.22 $\pm$ 0.12	97.31 $\pm$ 0.10	94.57 $\pm$ 0.01	92.79 $\pm$ 0.02	92.01 $\pm$ 0.04	
PAC-Momentum ($m = 0.9$)	85.28 $\pm$ 0.07	97.26 $\pm$ 0.12	95.68 $\pm$ 1.34	92.74 $\pm$ 0.45	89.83 $\pm$ 0.05	89.12 $\pm$ 0.13	97.25 $\pm$ 0.14	94.44 $\pm$ 0.04	92.66 $\pm$ 0.00	91.87 $\pm$ 0.05	
PAC-Momentum ($m = 0.1$)	85.15 $\pm$ 0.10	97.46 $\pm$ 0.23	96.81 $\pm$ 0.63	93.14 $\pm$ 0.11	90.08 $\pm$ 0.02	89.21 $\pm$ 0.10	97.32 $\pm$ 0.11	94.60 $\pm$ 0.05	92.80 $\pm$ 0.01	92.00 $\pm$ 0.03	
PAC-Residual (Interval)	85.16 $\pm$ 0.09	97.46 $\pm$ 0.23	96.86 $\pm$ 0.63	93.16 $\pm$ 0.11	90.08 $\pm$ 0.12	89.21 $\pm$ 0.14	97.31 $\pm$ 0.10	94.56 $\pm$ 0.04	92.79 $\pm$ 0.04	92.01 $\pm$ 0.04	
PAC-Residual (Cross)	85.15 $\pm$ 0.10	97.46 $\pm$ 0.23	96.90 $\pm$ 0.63	93.17 $\pm$ 0.12	90.06 $\pm$ 0.08	89.22 $\pm$ 0.13	97.31 $\pm$ 0.10	94.57 $\pm$ 0.04	92.79 $\pm$ 0.02	92.01 $\pm$ 0.03	

C. Prototype-Anchor Calibration Strategy

Table XIII compares DSSG-PAC with momentum- and residual-based calibration variants. At each periodic calibration event, DSSG-PAC updates the prototype cache with the current text-encoded class prototypes, i.e., $P_j^{\text{cache}} \leftarrow P_j^{\text{cur}}$, and keeps it fixed until the next calibration. PAC-Momentum introduces temporal smoothing as

$$P_j^{\text{cache}} \leftarrow \text{Norm}(mP_{j-1}^{\text{cache}} + (1-m)P_j^{\text{cur}}), \quad (19)$$

while the residual variants use

$$\tilde{P}_j = \text{Norm}(P_j^{\text{cache}} + \Delta P_j). \quad (20)$$

PAC-Residual (Interval) resets ΔP_j after each calibration, whereas PAC-Residual (Cross) retains it across calibration intervals.

As shown in Table XIII, strong momentum smoothing ($m = 0.9$) degrades performance, while weak smoothing ($m = 0.1$) and the residual variants provide only marginal and inconsistent changes relative to DSSG-PAC. These results indicate that the periodic calibration in DSSG-PAC is sufficient to track the evolving text representation space without additional smoothing or residual correction.

D. Sensitivity Analysis of Hyper-parameter λ_3

To avoid task-specific hyperparameter tuning, we follow the default settings of ImCapSFDA for λ_1 and λ_2 without additional search. We then vary only the DSSG-specific λ_3 for sensitivity analysis. As shown in Fig. 7, DSSG maintains

stable performance across a broad range of λ_3 values while consistently outperforming the ImCapSFDA baseline, indicating low sensitivity to λ_3 .

E. Proofs for the SFF-DA Risk Analysis

This appendix proves Theorem 1 and Corollary 1 and states precisely the conditions under which the population certificate can be estimated from finite unlabeled data.

1) *Notation and tie convention:* For $p, q \in \Delta^{K-1}$, let $h_p(x)$ and $h_q(x)$ be top-1 classifiers under the same deterministic tie-breaking rule. Write $q_{(1)}(x) \geq q_{(2)}(x)$ for the two largest coordinates of $q(x)$ and $m_q(x) = q_{(1)}(x) - q_{(2)}(x)$. If $m_q(x) = 0$, Eq. (14) assigns $\Psi(p, q; x) = 1$; hence tied teacher predictions are conservatively treated as uncanceled.

2) *A pointwise margin-discrepancy certificate:* **Lemma 1.** For every x ,

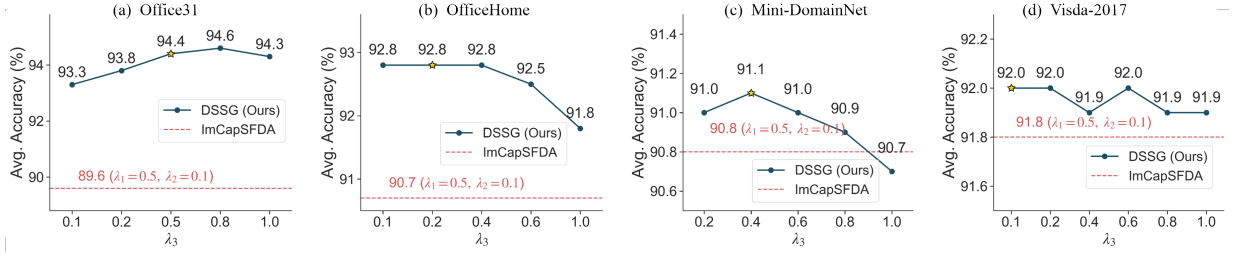
$$1[h_p(x) \neq h_q(x)] \leq \Psi(p, q; x). \quad (21)$$

Proof. If $h_p(x) = h_q(x)$, the left-hand side is zero and the claim is immediate. Otherwise, let $a = h_q(x)$ and $j = h_p(x) \neq a$. Since j is selected by the student, $p_j \geq p_a$. Since a is selected by the teacher,

$$q_a - q_j \geq q_{(1)} - q_{(2)} = m_q. \quad (22)$$

Consequently,

$$(p_j - q_j) - (p_a - q_a) = (p_j - p_a) + (q_a - q_j) \geq m_q. \quad (23)$$

Fig. 7. Hyper-Parameter Sensitivity Analysis of λ_3 on ViT-B/16

For arbitrary real numbers u, v , $u^2 + v^2 \geq (v - u)^2 / 2$. Applying this inequality to $u = p_a - q_a$ and $v = p_j - q_j$, and using Eq. (23), gives

$$\|p - q\|_2^2 \geq (p_a - q_a)^2 + (p_j - q_j)^2 \quad (24)$$

$$\geq \frac{m_q^2}{2}. \quad (25)$$

Thus, whenever $m_q > 0$ and $h_p \neq h_q$,

$$\frac{2\|p - q\|_2^2}{m_q^2} \geq 1, \quad (26)$$

so $\Psi(p, q; x) = 1$. If $m_q = 0$, Ψ equals one by definition. This proves Eq. (21). $\square$

Taking the expectation of Eq. (21) over Q_X immediately yields

$$d_Q(h_p, h_q) := \Pr_{Q_X}[h_p(x) \neq h_q(x)] \leq C_Q(p, q). \quad (27)$$

3) *Target-risk sandwich:* **Lemma 2.** For any two classifiers h_p, h_q under the multiclass 0–1 loss,

$$|R_Q(h_p) - R_Q(h_q)| \leq d_Q(h_p, h_q). \quad (28)$$

Proof. For every (x, y) ,

$$|1[h_p(x) \neq y] - 1[h_q(x) \neq y]| \leq 1[h_p(x) \neq h_q(x)]. \quad (29)$$

Taking expectations and applying $|\mathbb{E}Z| \leq \mathbb{E}|Z|$ proves Eq. (28). $\square$

Since $R_Q(h_q) = \eta_Q$, Lemma 2 and Eq. (27) imply

$$\eta_Q - C_Q(p, q) \leq R_Q(h_p) \leq \eta_Q + C_Q(p, q). \quad (30)$$

Intersecting this interval with $[0, 1]$ proves Theorem 1. Notice that only an upper bound on $d_Q(h_p, h_q)$ is available. Therefore, the generally invalid expression $|\eta_Q - C_Q|$ must not be used as a lower bound.

4) *Teacher-risk uncertainty interval:* If external calibration, an independently labeled diagnostic set, or an explicitly stated assumption provides

$$\underline{\eta} \leq \eta_Q \leq \bar{\eta}, \quad (31)$$

then Theorem 1 implies the robust interval

$$\max\{0, \underline{\eta} - C_Q(p, q)\} \leq R_Q(h_p) \leq \min\{1, \bar{\eta} + C_Q(p, q)\}. \quad (32)$$

The interval $[\underline{\eta}, \bar{\eta}]$ is not identifiable from Q_X alone: the same unlabeled marginal may be paired with different target labeling functions. Accordingly, Eq. (32) must not be described as fully observable in a label-free training protocol. In our analysis, target labels may be used only for post-hoc oracle evaluation of η_Q , never for adaptation or model selection.

5) *Finite-sample certificate:* Let $S_U = \{x_i\}_{i=1}^n$ be an i.i.d. unlabeled sample from Q_X that is independent of all training, hyperparameter selection, and stopping decisions used to obtain p and q . Define

$$\hat{C}_{S_U}(p, q) = \frac{1}{n} \sum_{i=1}^n \Psi(p, q; x_i). \quad (33)$$

Because $0 \leq \Psi \leq 1$, Hoeffding's inequality gives, with probability at least $1 - \delta$,

$$C_Q(p, q) \leq \hat{C}_{S_U}(p, q) + \sqrt{\frac{\ln(1/\delta)}{2n}}. \quad (34)$$

Combining Eqs. (32) and (34) yields

$$R_Q(h_p) \leq \min\left\{1, \bar{\eta} + \hat{C}_{S_U}(p, q) + \sqrt{\frac{\ln(1/\delta)}{2n}}\right\}. \quad (35)$$

If the same transductive target sample is used both to train the models and to compute $\hat{C}$, the independence condition is violated and Eq. (34) does not follow from Hoeffding's inequality. In that case, $\hat{C}$ is reported only as an empirical diagnostic unless cross-fitting, a uniform-convergence argument, or a PAC-Bayes analysis is introduced.

6) *Relation to Dynamic CMKD:* Let $p = P_{\text{stu}}$, $q = P_{\text{tea}}^*$, $c = \exp[-D_{\text{KL}}(p||q)]$, and $G(r) = 1 - \|r\|_2^2$. Since

$$G\left(\frac{p+q}{2}\right) = \frac{G(p) + G(q)}{2} + \frac{1}{4}\|p - q\|_2^2, \quad (36)$$

Eq. (9) of the main paper can be rewritten exactly as

$$\begin{aligned} \mathcal{L}_{\text{cmkd}}^* &= \frac{\alpha(1+c)}{2}G(p) + \left[\frac{\alpha(1-c)}{2} + \beta\right]G(q) \\ &\quad + \frac{\alpha(1-c)}{4}\|p - q\|_2^2. \end{aligned} \quad (37)$$

Thus, Dynamic CMKD contains an explicit consistency penalty whose coefficient increases monotonically with $D_{\text{KL}}(p||q)$. This establishes alignment between the training objective and the distribution-discrepancy component of Eq. (14). It does not, by itself, prove monotonic decrease of the unweighted quantity $\mathbb{E}\|p - q\|_2^2$, because the coefficient and the other Gini terms co-evolve during optimization.

Likewise, Theorem 1 does not prove that the class-anchor stream reduces η_Q . A high teacher margin represents low predictive ambiguity, not semantic correctness. Any claim that class anchors improve teacher correctness must be supported empirically, for example by oracle teacher pseudo-label accuracy that is used only for analysis.

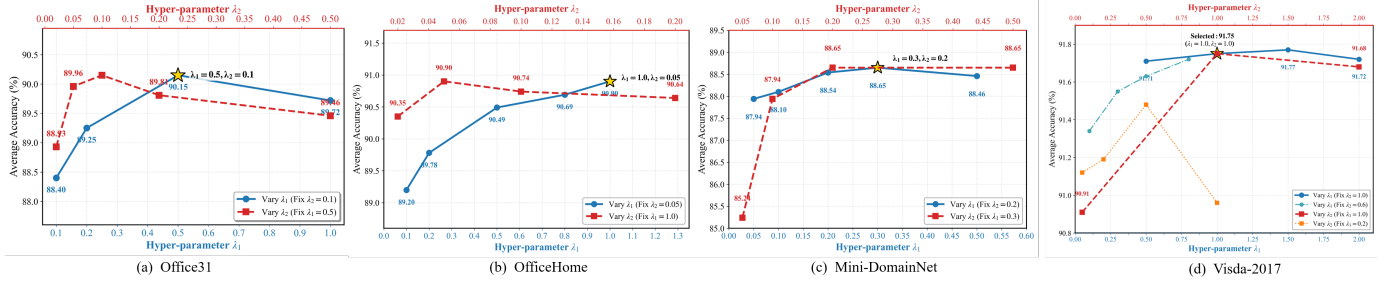


Fig. 8. Sensitivity of the ImCapFusionSFDA baseline to λ_1 (blue solid) and λ_2 (red dashed). The star ($\star$) indicates the selected optimal configuration.



Fig. 9. Grad-CAM visualization samples.

VII. SUPPLEMENTARY MATERIAL

A. Optimization of Baseline Hyper-parameters λ_1 and λ_2

Although ImCapSFDA reports benchmark results, its optimal λ_1 and λ_2 settings are not provided. We therefore optimize these hyperparameters for the reproduced baseline, as shown in Fig. 8, and directly inherit the selected settings for DSSG without additional tuning.

B. Activation Map Visualization

As shown in Fig. 9, DSSG exhibits more concentrated activations on discriminative object regions and reduced responses to background regions compared with the baseline, indicating improved task-relevant semantic focus.